Chapter 2

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Problem 1.

Let Ω be a finite set. Show that the set of all subsets of Ω, $\mathscr{P}(\Omega)$, is also finite and that it is a σ-algebra.

Solution.

Since Ω is finite, let $\Omega = \{\omega_1, ..., \omega_n\}$. Any subset of Ω can be formed by considering every element of Ω and either including it or excluding it in the subset. That is, for every element ω_i, $i = 1, ..., n$, we either include ω_i in the subset or exclude it, and so there are exactly two choices for every element. Therefore, the total number of possible subsets is 2^n, i.e., $\mathscr{P}(\Omega)$ is also finite.

We now show that $\mathscr{P}(\Omega)$ is a σ-algebra. Since $\varnothing, \Omega \subseteq \Omega$, we have that $\varnothing, \Omega \in \mathscr{P}(\Omega)$. Also, if $A \in \mathscr{P}(\Omega)$, then $A \subseteq \Omega$ and so $A^c = \Omega \setminus A \subseteq \Omega$, that is, $A^c \in \mathscr{P}(\Omega)$. Lastly, let $\{A_j\}_{j\geq 1}$ be a sequence of subsets of Ω. Then, for any $x \in \bigcup_{j\geq 1} A_j$, there exists some j_0 such that $x \in A_{j_0}$. But since $A_{j_0} \subseteq \Omega$, $x \in \Omega$, and so $\bigcup_{j\geq 1} A_j \subseteq \Omega$. Thus, $\bigcup_{j\geq 1} A_j \in \mathscr{P}(\Omega)$. $\square$

Problem 2.

Let $(\mathscr{G}_\alpha)_{\alpha\in A}$ be an arbitrary family of σ-algebras defined on an abstract space Ω. Show that $\mathscr{H} = \bigcap_{\alpha\in A} \mathscr{G}_\alpha$ is also a σ-algebra.

Solution.

Since $\mathscr{G}_\alpha$ is a σ-algebra for all $\alpha \in A$, $\varnothing, \Omega \in \mathscr{G}_\alpha$, and so $\varnothing, \Omega \in \bigcap_{\alpha\in A} \mathscr{G}_\alpha = \mathscr{H}$.

Let $A \in \mathscr{H}$. Then, for all $\alpha \in A$, $A \in \mathscr{G}_\alpha$, and so $A^c \in \mathscr{G}_\alpha$, for $\mathscr{G}_\alpha$ is a σ-algebra. Hence, $A^c \in \bigcap_{\alpha\in A} \mathscr{G}_\alpha = \mathscr{H}$.

Lastly, let $(A_j)_{j\geq 1}$ be a sequence of sets in $\mathscr{H}$. Then, for all $j \geq 1$, $A_j \in \mathscr{G}_\alpha$ for all $\alpha \in A$. Since $\mathscr{G}_\alpha$ is a σ-algebra, it is closed under countable unions. Hence, $\bigcup_{j\geq 1} A_j \in \mathscr{G}_\alpha$ for all $\alpha \in A$, and therefore $\bigcup_{j\geq 1} A_j \in \bigcap_{\alpha\in A} \mathscr{G}_\alpha = \mathscr{H}$. Thus, $\mathscr{H}$ is a σ-algebra. $\square$

Problem 3.

Let $(A_n)_{n\geq 1}$ be a sequence of sets. Show that (De Morgan's Laws)

1. $\left(\bigcup_{n\geq 1} A_n\right)^c = \bigcap_{n\geq 1} A_n^c$
2. $\left(\bigcap_{n\geq 1} A_n\right)^c = \bigcup_{n\geq 1} A_n^c$.

Solution.

1. Let $x \in \left(\bigcup_{n\geq 1} A_n\right)^c$. Then, $x \notin \bigcup_{n\geq 1} A_n$; that is, for all $n \in \mathbb{N}$, $x \notin A_n$. Hence, $x \in A_n^c$ for all $n \in \mathbb{N}$, and so $x \in \bigcap_{n\geq 1} A_n^c$.

   Now suppose $x \in \bigcap_{n\geq 1} A_n^c$. Then, for all $x \in \mathbb{N}$, $x \in A_n^c$, that is, $x \notin A_n$ for any $n \in \mathbb{N}$. Therefore, $x \notin \bigcup_{n\geq 1} A_n$, and so $x \in \left(\bigcup_{n\geq 1} A_n\right)^c$.

2. Let $x \in \left(\bigcap_{n\geq 1} A_n\right)^c$. Then, $x \notin \bigcap_{n\geq 1} A_n$, that is, there exists some $n_0 \in \mathbb{N}$ such that $x \notin A_{n_0}$. Therefore, $x \in A_{n_0}^c$, and so $x \in \bigcup_{n\geq 1} A_n^c$.

   Now suppose $x \in \bigcup_{n\geq 1} A_n^c$. Then, for some $n_0 \in \mathbb{N}$, $x \in A_{n_0}^c$, that is, $x \notin A_{n_0}$. Hence, $x \notin \bigcap_{n\geq 1} A_n$, and so $x \in \left(\bigcap_{n\geq 1} A_n\right)^c$.

$\square$

Problem 4.

Let $\mathscr{A}$ be a σ-algebra and $(A_n)_{n\geq 1}$ a sequence of events in $\mathscr{A}$. Show that

1. $\liminf_{n\to\infty} A_n \in \mathscr{A}$;
2. $\limsup_{n\to\infty} A_n \in \mathscr{A}$; and
3. $\liminf_{n\to\infty} A_n \subseteq \limsup_{n\to\infty} A_n$.

Solution.

Since

$$\limsup_{n\to\infty} A_n = \bigcap_{n\geq 1} \bigcup_{m\geq n} A_m$$

and

$$\liminf_{n\to\infty} A_n = \bigcup_{n\geq 1} \bigcap_{m\geq n} A_m,$$

and since a σ-algebra is closed under countable unions and countable intersections, $\limsup_{n\to\infty} A_n \in \mathscr{A}$ and $\liminf_{n\to\infty} A_n \in \mathscr{A}$.

Let $x \in \liminf_{n\to\infty} A_n = \bigcup_{n\geq 1} \bigcap_{m\geq n} A_m$. Then, for some $n_0 \in \mathbb{N}$, $x \in \bigcap_{m\geq n_0} A_m$. Hence, for all $m \geq n_0$, $x \in A_m$. But then, $x \in \bigcup_{m\geq n} A_m$ for all $n \geq 1$, and therefore $x \in \bigcap_{n\geq 1} \bigcup_{m\geq n} A_m = \limsup_{n\to\infty} A_n$, as required. $\square$

Problem 5.

Let $(A_n)_{n\geq 1}$ be a sequence of sets. Show that

$$\limsup_{n\to\infty} 1_{A_n} - \liminf_{n\to\infty} 1_{A_n} = 1_{\{\limsup_n A_n \setminus \liminf_n A_n\}}$$

(where $A \setminus B = A \cap B^c$ whenever $B \subseteq A$).

Solution.

Since $\liminf_n 1_{A_n} = \lim_{n\to\infty} \inf\{1_{A_n}, 1_{A_{n+1}}, ...\}$, it is either 1 or 0. Consider the case when it is 1. Then, $\limsup_n 1_{A_n} = \lim_{n\to\infty} \sup\{1_{A_n}, 1_{A_{n+1}}, ...\} = 1$ since for any n, $\inf\{1_{A_n}, 1_{A_{n+1}}, ...\} \leq \sup\{1_{A_n}, 1_{A_{n+1}}, ...\}$. Let $x \in \Omega$, where Ω is the abstract space under consideration. The fact that $\liminf_n 1_{A_n} = 1$ implies that for large enough n, $1_{A_n}(x) = 1$, that is, $x \in A_n$. Hence, $x \in \bigcup_{n\geq 1}\bigcap_{m\geq n} A_m$, i.e., $x \in \liminf_n A_n$, and by Problem 4, $x \in \limsup_n A_n$. Therefore, $x \notin \limsup_n A_n \setminus \liminf_n A_n$, and so $1_{\{\limsup_n A_n \setminus \liminf_n A_n\}}(x) = 0 = \limsup_n 1_{A_n}(x) - \liminf_n 1_{A_n}(x)$.

Now suppose that $\liminf_n 1_{A_n} = 0$. Then, $\limsup_n 1_{A_n}$ is either 0 or 1. If it is 0, then $\lim_{n\to\infty} \sup\{1_{A_n}, 1_{A_{n+1}}, ...\} = 0$, that is, for large enough n, $1_{A_n}(x) = 0$, and so $x \notin A_n$. Hence, x is only in finitely many A_n, and so $x \notin \limsup_n A_n$, implying that $x \notin \limsup_n A_n \setminus \liminf_n A_n$, i.e., $1_{\{\limsup_n A_n \setminus \liminf_n A_n\}}(x) = 0 = \limsup_n 1_{A_n}(x) - \liminf_n 1_{A_n}(x)$.

If $\limsup_n 1_{A_n} = 1$ and $\liminf_n 1_{A_n} = 0$, then $\lim_{n\to\infty} \sup\{1_{A_n}, 1_{A_{n+1}}, ...\} = 1$, and so, for any $n \in \mathbb{N}$, there exists some $N \geq n$ such that $x \in A_N$. Thus, $x \in \bigcap_{n\geq 1}\bigcup_{m\geq n} A_m = \limsup_n A_n$. However, $\liminf_n 1_{A_n} = 0$, i.e., $\lim_{n\to\infty} \inf\{1_{A_n}, 1_{A_{n+1}}, ...\} = 0$, and so for any $n \in \mathbb{N}$, there exists $N' \geq n$ such that $x \notin A_{N'}$. Hence, $x \notin \bigcup_{n\geq 1}\bigcap_{m\geq n} A_m = \liminf_n A_n$, that is, $x \in \limsup_n A_n \setminus \liminf_n A_n$, and so $1_{\{\limsup_n A_n \setminus \liminf_n A_n\}}(x) = 1 = \limsup_n 1_{A_n}(x) - \liminf_n 1_{A_n}(x)$, which concludes the proof. $\square$

Problem 6.

Let $\mathscr{A}$ be a σ-algebra of subsets of Ω and let $B \in \mathscr{A}$. Show that $\mathscr{F} = \{A \cap B : A \in \mathscr{A}\}$ is a σ-algebra of subsets of B. Is it still true when B is a subset of Ω that does not belong to $\mathscr{A}$?

Solution.

We first observe that $A \cap B \subseteq B$ and so $\mathscr{F}$ is a collection of subsets of B. Since $\varnothing = \varnothing \cap B$ and $\varnothing \in \mathscr{A}$, for $\mathscr{A}$ is a σ-algebra, we have that $\varnothing \in \mathscr{F}$. Similarly, $B = \Omega \cap B$ and $\Omega \in \mathscr{A}$, so $B \in \mathscr{F}$.

We now show that $\mathscr{F}$ is closed under complements. Let $E \in \mathscr{F}$. Then, there exists some $A_0 \in \mathscr{A}$ such that $E = A_0 \cap B$, and so the complement of E in B is given by $B \setminus E = B \cap E^c = B \cap (A_0 \cap B)^c = B \cap (A_0^c \cup B^c) = B \cap A_0^c$. Now, $A_0 \in \mathscr{A}$ and so $A_0^c \in \mathscr{A}$. Hence, $B \setminus E \in \mathscr{F}$.

Lastly, we show that $\mathscr{F}$ is closed under countable unions. Let $(A_n)_{n\geq 1}$ be a sequence of sets in $\mathscr{F}$. Then, for every $n \geq 1$, there exists some $E_n \in \mathscr{A}$ such that $A_n = E_n \cap B$. Therefore, $\bigcup_{n\geq 1} A_n = \bigcup_{n\geq 1}(E_n \cap B) = B \cap \bigcup_{n\geq 1} E_n$, and since $\bigcup_{n\geq 1} E_n \in \mathscr{A}$, we have that $\left(\bigcup_{n\geq 1} E_n\right) \cap B \in \mathscr{F}$, as required, and so $\mathscr{F}$ is a σ-algebra.

Since the proof above does not depend on B being an element of $\mathscr{A}$, the assertion is also true when $B \subseteq \Omega$ is not an element of $\mathscr{A}$. $\square$

Problem 7.

Let f be a function mapping Ω to another space E with a σ-algebra $\mathscr{E}$. Let $\mathscr{A} = \{A \subseteq \Omega : \text{there exists } B \in \mathscr{E} \text{ with } A = f^{-1}(B)\}$. Show that $\mathscr{A}$ is a σ-algebra on Ω.

Solution.

We first show that $\varnothing, \Omega$ are in $\mathscr{A}$. Since $\mathscr{E}$ is a σ-algebra, $\varnothing \in \mathscr{E}$. Also, $f^{-1}(\varnothing) = \{\omega \in \Omega : f(\omega) \in \varnothing\} = \varnothing$ and so $\varnothing \in \mathscr{A}$. Further, $f^{-1}(E) = \Omega$ and $E \in \mathscr{E}$, that is, $\Omega \in \mathscr{A}$.

Now, we show that $\mathscr{A}$ is closed under complements. Let $A \in \mathscr{A}$. Then, there exists $B \in \mathscr{E}$ such that $A = f^{-1}(B) = \{\omega \in \Omega : f(\omega) \in B\}$, and so $A^c = \{\omega \in \Omega : f(\omega) \notin B\}$, which is the same as $\{\omega \in \Omega : f(\omega) \in B^c\} = f^{-1}(B^c)$, but $B^c \in \mathscr{E}$ since $B \in E$, and therefore $A^c \in \mathscr{A}$.

Lastly, we show that $\mathscr{A}$ is closed under countable unions. Let $(A_n)_{n\geq 1}$ be a sequence of sets in $\mathscr{A}$. Then, for every $n \geq 1$, there exists $B_n \in \mathscr{E}$ such that $A_n = f^{-1}(B_n) = \{\omega \in \Omega : f(\omega) \in B_n\}$, and so $\bigcup_{n\geq 1} A_n = \{\omega \in \Omega : f(\omega) \in \bigcup_{n\geq 1} B_n\} = f^{-1}\left(\bigcup_{n\geq 1} B_n\right)$, and since $\bigcup_{n\geq 1} B_n \in \mathscr{E}$, we have $\bigcup_{n\geq 1} A_n \in \mathscr{A}$, as desired. $\square$

Problem 8.

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function, and let $\mathscr{A} = \{A \subseteq \mathbb{R} : \exists B \in \mathscr{B} \text{ with } A = f^{-1}(B)\}$ where $\mathscr{B}$ are the Borel subsets of the range space $\mathbb{R}$. Show that $\mathscr{A} \subseteq \mathscr{B}$, the Borel subsets of the domain space $\mathbb{R}$.

For problems 9 to 15, we assume a fixed abstract space Ω, a σ-algebra $\mathscr{A}$, and a Probability P defined on $(\Omega, \mathscr{A})$. The sets A, B, A_i, etc... always belong to $\mathscr{A}$.

Problem 9.

For $A, B \in \mathscr{A}$ with $A \cap B = \varnothing$, show $P(A \cup B) = P(A) + P(B)$.

Solution.

P is a Probability measure and therefore is finitely additive. Hence, $P(A \cup B) = P(A) + P(B)$. $\square$

Problem 10.

For $A, B \in \mathscr{A}$, show $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Solution.

$A \cup B = A \cup (B \setminus A)$, and since A and $B \setminus A$ are disjoint, $P(A \cup B) = P(A) + P(B \setminus A)$. Now, $B \setminus A = B \setminus (A \cap B)$, and since $A \cap B \subseteq B$, $P(B \setminus A) = P(B) - P(A \cap B)$. Thus, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. $\square$

Problem 11.

For $A \in \mathscr{A}$, show $P(A) = 1 - P(A^c)$.

Solution.

Since $\Omega = A \cup A^c$, $P(\Omega) = P(A) + P(A^c)$, and so $P(A) = 1 - P(A^c)$. $\square$

Problem 12.

For $A, B \in \mathscr{A}$, show $P(A \cap B^c) = P(A) - P(A \cap B)$.

Solution.

$A \cap B^c = A \setminus (A \cap B)$ and $A \cap B \subseteq A$. Therefore, $P(A \cap B^c) = P(A \setminus (A \cap B)) = P(A) - P(A \cap B)$. $\square$

Problem 13.

Let $A_1, ..., A_n$ be given events. Show that

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_i P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k) - ... + (-1)^{n+1} P(A_1 \cap A_2 \cap ... \cap A_n)$$

where (for example) $\sum_{i<j}$ means to sum over all ordered pairs (i, j) with $i < j$.

Solution.

The statement is trivial for $n = 1$. Suppose it is true for $n-1$. Then,

$$\begin{aligned} P\left(\bigcup_{i=1}^{n} A_i\right) &= P\left(\bigcup_{i=1}^{n-1} A_i \cup A_n\right) \\ &= P\left(\bigcup_{i=1}^{n-1} A_i\right) + P(A_n) - P\left(\bigcup_{i=1}^{n-1} A_i \cap A_n\right). \end{aligned}$$

Now, $P\left(\bigcup_{i=1}^{n-1} A_i\right) = \sum_{i=1}^{n-1} P(A_i) - \sum_{1\le i<j\le n-1} P(A_i \cap A_j) + \ldots + (-1)^n P(A_1 \cap \ldots \cap A_{n-1})$. Further, let $B_i = A_i \cap A_n$. Then,

$$\begin{aligned} P\left(\bigcup_{i=1}^{n-1} A_i \cap A_n\right) &= P\left(\bigcup_{i=1}^{n-1} B_i\right) \\ &= \sum_{i=1}^{n-1} P(B_i) - \sum_{1\le i<j\le n-1} P(B_i \cap B_j) + \ldots \\ &\quad + (-1)^n P(B_1 \cap \ldots \cap B_{n-1}) \\ &= \sum_{i=1}^{n-1} P(A_i \cap A_n) - \sum_{1\le i<j\le n-1} P(A_i \cap A_j \cap A_n) + \ldots \\ &\quad + (-1)^n P(A_1 \cap \ldots \cap A_n). \end{aligned}$$

Observe that $\sum_{1\le i<j\le n-1} P(A_i \cap A_j)$ from $P\left(\bigcup_{i=1}^{n-1} A_i\right)$and $\sum_{i=1}^{n-1} P(A_i \cap A_n)$ from $P\left(\bigcup_{i=1}^{n-1} B_i\right)$ add to form $\sum_{1\le i<j\le n} P(A_i \cap A_j)$, and so on, as required. $\square$

Problem 14.

Suppose $P(A) = \frac{3}{4}$ and $P(B) = \frac{1}{3}$. Show that always $\frac{1}{12} \le P(A \cap B) \le \frac{1}{3}$.

Problem 15.

(Subadditivity) Let $A_i \in \mathscr{A}$ be a sequence of events. Show that

$$P\left(\bigcup_{i=1}^{n} A_i\right) \le \sum_{i=1}^{n} P(A_i),$$

each n, and also

$$P\left(\bigcup_{i\ge 1} A_i\right) \le \sum_{i\ge 1} P(A_i).$$

Solution.

Define $B_1 := A_1$, $B_2 = A_2 \setminus A_1$, $B_3 = A_3 \setminus (A_1 \cup A_2)$,..., $B_k = A_k \setminus \bigcup_{j=1}^{k-1} A_j$. Then, $(B_i)_{i=1}^{n}$ is a sequence of pairwise disjoint sets such that $\bigsqcup_{i=1}^{n} B_i = \bigcup_{i=1}^{n} A_i$. Since P is a probability measure, it is finitely additive and so $P\left(\bigcup_{i=1}^{n} A_i\right) = P\left(\bigsqcup_{i=1}^{n} B_i\right) = \sum_{i=1}^{n} P(B_i)$, but $B_i \subseteq A_i$ for all $i = 1, 2, ..., n$, due to which $P(B_i) \leq P(A_i)$. Hence, $\sum_{i=1}^{n} P(B_i) \leq \sum_{i=1}^{n} P(A_i)$.

Similarly, using countable additivity instead of finite additivity gives

$$P\left(\bigcup_{i\geq 1} A_i\right) = P\left(\bigsqcup_{i\geq 1} B_i\right) = \sum_{i\geq 1} P(B_i) \leq \sum_{i\geq 1} P(A_i).$$

□

Problem 16.

Suppose that Ω is an infinite set (countable or not), and let $\mathscr{A}$ be the family of all subsets which are either finite or have a finite complement. Show that $\mathscr{A}$ is an algebra, but not a σ-algebra.

Solution.

$\varnothing$ is a finite subset of Ω and so $\varnothing \in \Omega$. Further, $\Omega^c = \varnothing$, and so $\Omega \in \mathscr{A}$.

Let $A \in \mathscr{A}$. Then, either A is finite or A^c is finite. In the former case, $A^c \in \Omega$ since the complement of A^c is A, which is finite; in the latter case, A^c itself is finite and so $A^c \in \Omega$.

Suppose $A_1, ..., A_n \in \mathscr{A}$. There are two possibilities: either A_j is finite for all $j = 1, ..., n$ or at least one of the A_j's is infinite. If all A_j's are finite, then $\bigcup_{j=1}^{n} A_j$ is also finite, since a finite union of finite sets is finite, and so $\bigcup_{j=1}^{n} A_j \in \mathscr{A}$. Now, suppose A_{j_0} is infinite, where $j_0 \in \{1, ..., n\}$. Then, since $A_{j_0} \in \mathscr{A}$, $A_{j_0}^c$ is finite. Consider $B = \bigcap_{j=1}^{n} A_j^c$. Clearly, $B \subseteq \Omega$. Also, $B \subseteq A_{j_0}^c$, but since the latter set is finite, B is also finite. Hence, $B \in \mathscr{A}$, and so $B^c \in \mathscr{A}$. But $B = \bigcap_{j=1}^{n} A_j^c = (\bigcup_{j=1}^{n} A_j)^c$ and therefore $B^c = \bigcup_{j=1}^{n} A_j$, i.e., $\bigcup_{j=1}^{n} A_j \in \mathscr{A}$. Thus, $\mathscr{A}$ is an algebra.

To show that $\mathscr{A}$ is not a σ-algebra, we first consider the case where Ω is countably infinite. Let $\Omega = \{\omega_1, \omega_2, ...\}$. Define $A_j := \{\omega_{2j}\}$ for all $j \geq 1$. Then, each A_j is finite and so $A_j \in \mathscr{A}$ for all $j \geq 1$, but $\bigcup_{j\geq 1} A_j = \{\omega_2, \omega_4, ...\}$, which in infinite. Further, $(\bigcup_{j\geq 1} A_j)^c = \{\omega_1, \omega_3, ...\}$, which is also infinite, and so $\bigcup_{j\geq 1} A_j \notin \mathscr{A}$. Hence, $\mathscr{A}$ is not a σ-algebra.

If Ω is uncountably infinite, let $E = \{e_1, e_2, ...\}$ be a countably infinite subset of Ω (such a subset exists by the Axiom of Choice). Now, define $A_j := \{e_j\}$ for all $j \geq 1$. Each A_j is finite and so $A_j \in \mathscr{A}$ for all $j \geq 1$, but $\bigcup_{j\geq 1} A_j = E$, which is infinite, and E^c complement is also infinite. Therefore, $\bigcup_{j\geq 1} A_j \notin \mathscr{A}$, and so $\mathscr{A}$ is not a σ-algebra. □